

Urban Forecast: A Lightweight Forecasting Pipeline for Smart Waste Bin Monitoring in Smart Cities

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Summary

Urban Forecast is an open-source Python library that provides an end-to-end forecasting pipeline for smart waste bin monitoring using IoT sensor data.

Combining lightweight forecasting models, evaluation metrics, REST API integration, and optional LLM-based explanations, the software supports smart city applications and urban waste management optimization.

Designed with interpretability, simplicity, and ease of deployment in mind, Urban Forecast is suitable for both academic research and practical IoT environments.

This system was developed as part of an ongoing master's research project in the Programa de Pós-Graduação em Ciência da Computação (PPG-CC) at Universidade Federal de São Paulo (UNIFESP).

Its architecture integrates embedded devices based on ESP8266 microcontrollers ([Espressif Systems, 2025](#)) and ultrasonic sensors capable of monitoring waste levels in real time. Sensor measurements are transmitted to a Supabase database ([Supabase, 2026](#)) and processed by Urban Forecast to estimate future fill levels and support smart waste collection decisions.

These predictions can assist waste collection route optimization, reduce operational costs, prevent overflow events, and improve the allocation of collection resources.

Statement of need

Traditional urban waste collection systems commonly operate using static collection schedules, regardless of the actual fill level of waste bins. This approach often results in inefficient routes, unnecessary fuel consumption, increased operational costs, and delayed collection in high-demand areas ([Abdallah et al., 2019](#)).

Recent advances in IoT technologies enable the deployment of smart bins capable of continuously transmitting fill-level measurements. However, converting raw sensor measurements into actionable predictive insights remains a challenge for municipalities, researchers, and smart city initiatives.

Many forecasting approaches for IoT applications rely on machine learning models or deployment architectures that may introduce additional computational and operational complexity, potentially limiting their suitability for lightweight embedded monitoring systems ([Ahmed et al., 2022](#)).

Urban Forecast addresses this problem by providing an end-to-end forecasting framework specifically designed for smart waste monitoring. Unlike general-purpose forecasting libraries, the framework integrates IoT data ingestion, lightweight forecasting models, configurable

39 prediction pipelines, evaluation metrics, and REST API support within a unified architecture.
40 In addition, Urban Forecast can optionally generate natural language summaries of prediction
41 results using Large Language Models (LLMs) (Touvron et al., 2023).

42 The software is particularly relevant for applications in smart city research, urban analytics,
43 IoT monitoring systems, and waste management optimization. In addition, its accessible
44 architecture makes it suitable for educational and academic environments, where forecasting
45 methods and smart city technologies can be explored and evaluated.

46 Urban Forecast contributes to the field by offering a forecasting framework specifically tailored
47 to smart waste monitoring scenarios. The project emphasizes lightweight and interpretable
48 forecasting strategies that can operate in resource-constrained environments while integrating
49 IoT sensing, cloud-based data persistence, and AI-assisted explainability within a unified
50 architecture. This combination enables rapid experimentation, reproducible research, and
51 practical deployment in smart city applications.

52 State of the field

53 Smart waste management systems have become an important research topic within smart city
54 initiatives (Chaudhari et al., 2019). Several IoT-based architectures have been proposed for
55 monitoring urban waste containers using sensors, wireless communication, and cloud-based
56 infrastructures (Gharaibeh et al., 2017).

57 General-purpose forecasting libraries such as Prophet and StatsForecast provide powerful
58 forecasting capabilities for a wide range of applications. However, these libraries are not
59 specifically designed for smart waste monitoring workflows and typically require additional
60 integration layers for IoT data ingestion, REST APIs, or AI-assisted interpretation. Similarly,
61 IoT platforms such as ThingsBoard focus primarily on device connectivity and dashboard
62 visualization, offering limited forecasting support.

63 Machine learning and time-series forecasting approaches have been explored in urban monitoring
64 applications, but many solutions require complex infrastructures or rely on computationally
65 expensive models that are difficult to deploy in resource-constrained environments.

66 Urban Forecast complements these existing tools by providing a lightweight and modular
67 forecasting pipeline specifically designed for smart waste monitoring scenarios. The framework
68 emphasizes interpretability, simplicity, and ease of integration with cloud-based architectures
69 such as Supabase, enabling rapid experimentation and practical deployment in IoT environments
70 with limited computational resources.

71 Software design

72 Urban Forecast was designed using a modular architecture to facilitate experimentation with
73 different forecasting strategies and integration with external systems.

74 The software receives historical time-series data, performs data preparation and forecasting,
75 evaluates prediction quality using standard forecasting metrics, and can optionally generate
76 natural language summaries through LLM integration.

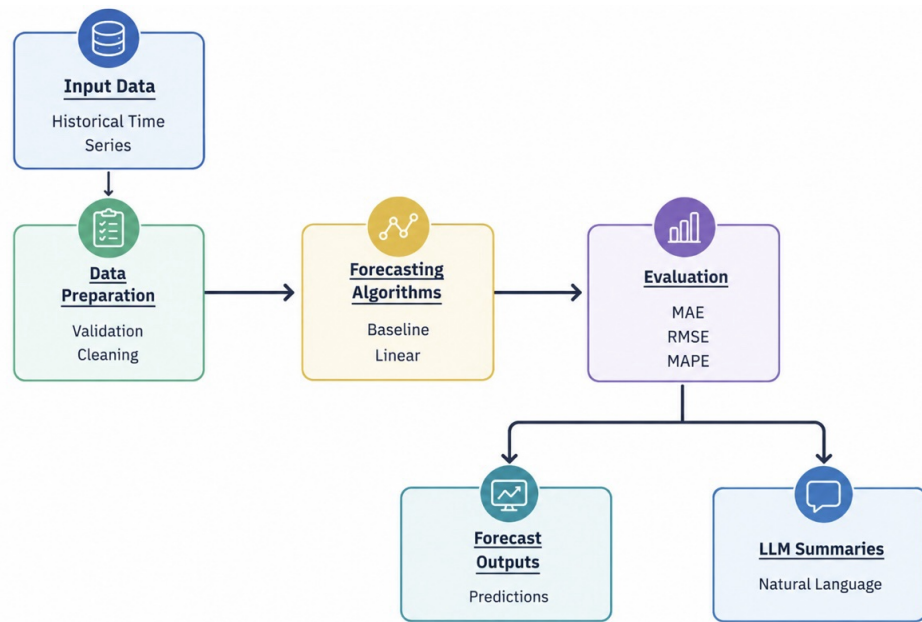


Figure 1: Architecture of the Urban Forecast framework. Historical time-series data are processed through forecasting and evaluation modules, producing prediction outputs and optional LLM-based summaries.

The framework is organized into modular components:

- `baseline.py`: heuristic forecasting model;
- `regression.py`: linear regression forecasting model;
- `pipeline.py`: unified execution pipeline;
- `ia.py`: LLM-based explanation generation;
- `api/`: REST API integration layer.

Urban Forecast currently implements two lightweight forecasting approaches designed for smart waste monitoring scenarios. The selected methods represent different forecasting strategies while maintaining low computational requirements and straightforward interpretability, characteristics that are particularly important in IoT-based environments.

The forecasting workflow consists of collecting fill-level measurements from ultrasonic sensors, storing the data in Supabase, preprocessing historical observations, applying forecasting models, and optionally generating natural language summaries through LLM integration.

Baseline Average Rate Model

The Baseline Average Rate model is a heuristic forecasting approach that estimates future fill levels based on the recent rate of change observed in historical measurements. The method assumes that the short-term filling behavior of a waste bin can be approximated by its most recent filling trend, making it suitable for scenarios where computational simplicity and rapid execution are required (Hyndman & Athanasopoulos, 2021).

The model estimates the recent filling rate using the median of observed rate-of-change values within a 96-hour window, reducing sensitivity to sensor noise and isolated outlier readings while preserving the model's simplicity and interpretability. A wider window was adopted after preliminary experiments showed that a narrower 48-hour window often contained too few observations to produce a stable rate estimate, particularly for sensors with longer sampling intervals.

Because the model relies on straightforward calculations rather than parameter optimization

or model training, it presents a low computational overhead and can be applied in resource-constrained environments. In addition, its transparent formulation facilitates interpretation of the generated forecasts and supports practical deployment in IoT-based monitoring systems.

Linear Regression Model

The Linear Regression model estimates future fill levels by fitting a linear relationship between time and observed fill percentages (James et al., 2021). This approach captures long-term filling tendencies and provides a simple statistical framework for forecasting when waste accumulation follows an approximately linear trend.

Compared to purely heuristic approaches, linear regression can produce smoother forecasts and reduce the influence of short-term fluctuations or noisy sensor measurements. The implementation relies on a robust regression estimator (Huber loss), which limits the influence of large isolated deviations, such as those introduced by sporadic waste-disposal events, while retaining the simplicity and computational efficiency of a linear model (Huber, 1964). This makes it suitable for exploratory forecasting studies and smart waste monitoring applications.

Evaluation metrics

Urban Forecast supports commonly used forecasting metrics for evaluating prediction quality and comparing forecasting strategies.

The current implementation supports:

- Mean Absolute Error (MAE), which measures the average magnitude of prediction errors (Willmott & Matsuura, 2005);
- Root Mean Squared Error (RMSE), which penalizes larger prediction deviations more strongly (Chai & Draxler, 2014);
- Mean Absolute Percentage Error (MAPE), which measures prediction error relative to the observed values.

These metrics allow quantitative evaluation of forecasting accuracy in smart waste monitoring scenarios and support comparison between heuristic and regression-based approaches (Hyndman & Koehler, 2006).

Experimental evaluation

The experimental dataset contains 180 observations generated from three simulated smart waste bins based on fill-level dynamics observed in the associated IoT monitoring system based on ESP8266 microcontrollers and ultrasonic sensors, with 60 measurements per sensor.

Historical sensor measurements were retrieved from the Supabase-based data infrastructure used in the associated research project. To better approximate real-world fill-level dynamics, the dataset incorporates a non-linear filling trend, moderate measurement noise, and sporadic discrete waste-disposal events that produce localized jumps in fill level. This evaluation setup is intended to avoid overly idealized linear dynamics while preserving an underlying filling tendency that can be exploited by both forecasting approaches.

The evaluation focused on estimating the remaining time until waste bins reached a predefined fill threshold. For each sensor, predictions were generated using a walk-forward scheme: at each step, the model was trained on all observations up to a given point and evaluated against the next time at which the fill level crossed the threshold. Predictions corresponding to remaining times greater than 72 hours were discarded to avoid unstable long-horizon estimates.

The forecasting approaches were evaluated using the metrics supported by Urban Forecast, including MAE, RMSE, and MAPE (Hyndman & Koehler, 2006). MAPE values are reported

147 as decimal fractions.

148 Table 1 summarizes the global forecasting performance of the evaluated models, including the
149 number of evaluated predictions (N) per model. The results show that the regression-based
150 approach achieved lower errors across all evaluation metrics.

Model	N	MAE	RMSE	MAPE
Baseline	78	7.966	10.328	0.243
Regression	78	2.503	3.489	0.108

151 The evaluated forecasting approaches presented different performance levels during the
152 experimental evaluation. The regression-based approach achieved substantially lower errors
153 compared to the heuristic baseline, reducing MAE from 7.966 hours to 2.503 hours and RMSE
154 from 10.328 hours to 3.489 hours.

155 The obtained performance suggests that modeling the temporal evolution of the fill level
156 provides more accurate remaining-time estimates than relying only on the median historical
157 filling rate.

158 This evaluation demonstrates that lightweight forecasting approaches can provide accurate
159 predictive estimates for smart waste monitoring scenarios while avoiding computationally
160 expensive models and complex deployment infrastructures.

161 Table 2 presents the forecasting performance across individual monitored sensors, including the
162 number of evaluated predictions (N) per sensor and model.

Sensor	Model	N	MAE	RMSE	MAPE
A1	Baseline	36	6.994	8.439	0.210
A1	Regression	36	2.754	4.128	0.078
A2	Baseline	24	5.121	6.012	0.191
A2	Regression	24	2.253	2.687	0.122
A3	Baseline	18	13.702	16.480	0.377
A3	Regression	18	2.335	3.010	0.147

163 The per-sensor evaluation confirms the consistent advantage of the regression-based approach
164 across all monitored waste bins. Sensor A3 exhibited the highest baseline error among the three
165 sensors, suggesting that its underlying fill-level dynamics were comparatively less amenable
166 to the simple rate-based heuristic, while the regression-based approach remained robust to
167 this variation. Differences in N across sensors arise from the walk-forward evaluation scheme
168 combined with the 72-hour horizon limit: sensors with longer sampling intervals produce fewer
169 valid evaluation points within this horizon.

170 Figure 2 presents the relationship between predicted and real remaining time values for both
171 evaluated forecasting approaches.

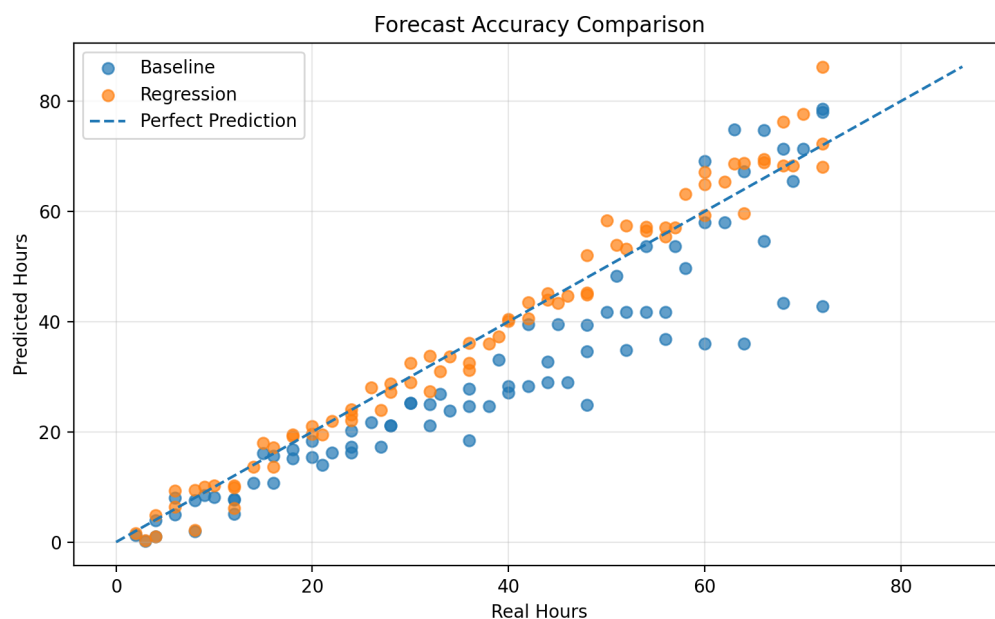


Figure 2: Comparison between predicted and real remaining time estimates for the evaluated forecasting approaches.

172 The regression-based approach presented predictions closer to the ideal prediction line, indicating
 173 better agreement between estimated and observed remaining time values under the evaluated
 174 monitoring conditions.

175 Figure 3 presents the mean absolute error (MAE) of both forecasting approaches as a function
 176 of the training window size, i.e., the number of historical observations available at the time of
 177 prediction. The horizontal reference line at $MAE = 0$ represents a perfect prediction scenario,
 178 where the estimated remaining time exactly matches the observed threshold-crossing time.

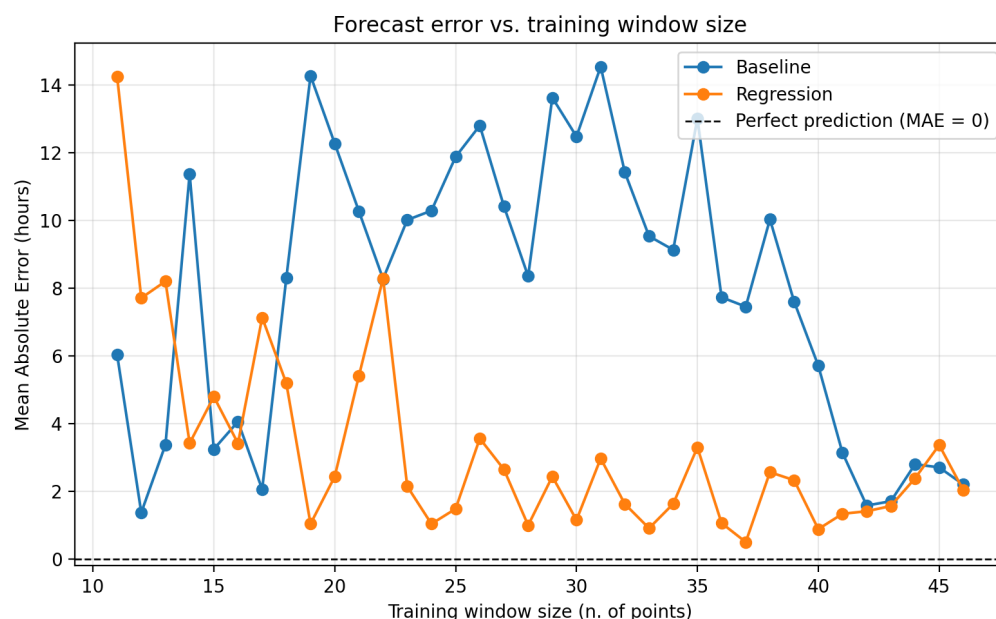


Figure 3: Mean absolute error of the evaluated forecasting approaches as a function of the training window size.

179 The distance from the curves to the $MAE = 0$ reference line indicates the magnitude of
 180 the forecasting error, with lower values corresponding to more accurate predictions. The
 181 regression-based approach converges to low and stable error values (typically below 5 hours)
 182 once approximately 20 training observations are available, whereas the baseline approach
 183 remains considerably more volatile across a wider range of training window sizes, with errors
 184 occasionally exceeding 15 hours. Both approaches converge to similarly low error levels once
 185 the training window approaches the full sensor history, suggesting that the regression model
 186 reaches a stable predictive regime earlier than the baseline model. This pattern is consistent
 187 with the per-sensor results, where the baseline model's accuracy is more strongly affected by
 188 sensor-specific fill-level dynamics, while the regression model maintains comparatively stable
 189 error levels across all evaluated sensors.

190 Additional experimental artifacts, including generated CSV reports, benchmark scripts, and
 191 visualization outputs, are publicly available in the project repository to support reproducibility
 192 and transparency.

193 AI-assisted explainability

194 Urban Forecast optionally integrates Large Language Models (LLMs) through the Groq API to
 195 automatically generate natural language summaries of the forecasting results. This capability
 196 translates numerical outputs, forecast trends, and estimated filling trajectories into concise
 197 textual explanations that can be more easily interpreted by non-technical users.

198 The explainability layer is designed to complement quantitative forecasting metrics rather than
 199 replace them. By providing human-readable descriptions of predicted waste accumulation
 200 patterns, the system can support operational decision-making and facilitate communication
 201 between technical and administrative stakeholders.

202 This functionality is particularly useful in smart city environments, where waste management
 203 operators may benefit from rapid interpretation of forecasting outputs without requiring
 204 expertise in time-series analysis or statistical modeling. The integration leverages the Groq

platform for low-latency inference and access to modern LLMs, enabling real-time generation of explanatory summaries (Groq, Inc., 2024).

Research impact statement

Urban Forecast is currently being used as part of an ongoing master's research project focused on smart waste collection systems based on IoT infrastructure and cloud computing.

The project contributes to smart city research by combining embedded IoT sensing, cloud-based data persistence, lightweight forecasting models, and AI-assisted explainability within a unified framework.

The modular design allows the software to be adapted for educational purposes, research experiments, and real-world urban monitoring systems.

Example usage

```
from urban_forecast.pipeline import run_pipeline
import pandas as pd

df = pd.DataFrame({
    "created_at": pd.date_range(
        start="2024-01-01",
        periods=10,
        freq="h"
    ),
    "fill_percent": [10,15,20,25,30,35,40,45,50,55]
})

result = run_pipeline(df)

print(result)
```

AI usage disclosure

All technical content, software implementation details, experimental results, and project information presented in this manuscript were produced, reviewed, and validated by the authors.

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